

SIMATS ENGINEERING
ASSESSMENT TOOL 3: MCQ Test

COURSE NAME:	Cloud Computing and Big Data Analytics	COURSE CODE:	CSA15
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COURSE OUTCOME COVERED

CO3: *Implement cloud-based solutions using cloud storage, web APIs, and platform services for real-world applications. (BL3)*

Assessment Tool Weightage: 10%

Dave's Taxonomy: Precision (Level 3)
Question Count: 15

Total Marks: 30

Code of Conduct

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: _____

Assessment Tasks

MCQ Test

1. Which cloud storage model is best suited for storing virtual machine disks?
 - A. Object storage
 - B. Block storage
 - C. File storage
 - D. Archive storage
2. Which protocol style is widely used for lightweight web APIs in cloud platforms?
 - A. SOAP only
 - B. REST
 - C. FTP
 - D. SMTP
3. Serverless computing is best described as:
 - A. Computing without networks
 - B. Running code without managing servers directly
 - C. Local execution only
 - D. Batch-only processing
4. Edge computing primarily helps by:
 - A. Increasing central storage only
 - B. Reducing latency near data sources
 - C. Eliminating internet access
 - D. Replacing cloud storage fully
5. Fog computing differs from edge computing because it:
 - A. Always runs inside sensors
 - B. Introduces an intermediate distributed layer between edge and cloud

- C. Works only in private clouds
 - D. Uses no network resources
6. Which storage model is commonly used for shared departmental files?
- A. File storage
 - B. Register storage
 - C. Cache-only storage
 - D. Tape-only storage
7. A browser calling a cloud web application typically communicates through:
- A. BIOS interrupts
 - B. Web APIs
 - C. Only command-line tools
 - D. VPN tunnels only
8. SAN stands for:
- A. Secure Access Node
 - B. Storage Area Network
 - C. Shared Application Namespace
 - D. Service Access Network
9. Which of the following is a common benefit of serverless architecture?
- A. Manual scaling of servers
 - B. Pay-per-use execution model
 - C. Dedicated physical hardware ownership
 - D. No monitoring required
10. Collaboration within the cloud usually improves:
- A. Shared access to services and documents
 - B. Hardware disposal speed
 - C. Cable management
 - D. BIOS configuration
11. Which one is an example of a platform service?
- A. Managed application runtime
 - B. Keyboard driver
 - C. Physical rack switch only
 - D. Local USB storage
12. Object storage is generally preferred for:
- A. Mounting OS boot volumes
 - B. Large unstructured cloud data such as images and backups
 - C. CPU scheduling
 - D. Hypervisor installation
13. Which component allows applications to communicate programmatically with cloud services?
- A. Heat sink
 - B. Web API
 - C. Mouse driver
 - D. Compiler cache
14. A low-latency smart factory use case is best supported by:
- A. Edge computing
 - B. Cold archive only
 - C. Offline storage
 - D. Manual processing
15. Which one is not a typical cloud access method?
- A. Browser-based access
 - B. API access
 - C. Platform console access
 - D. Paper-based access

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Conceptual Understanding	30%	Answers reflect strong and accurate concept mastery	Mostly accurate understanding	Partial understanding	Weak understanding
Accuracy of Responses	30%	Nearly all responses are correct	Most responses are correct	Some responses are correct	Few responses are correct
Application Awareness	15%	Shows strong recognition of practical cloud use cases	Reasonable recognition	Limited recognition	Weak recognition
Time Management	10%	Completes assessment efficiently	Completes on time	Needs extra time	Unable to finish well
Question Interpretation	15%	Interprets all items correctly	Misreads very few items	Misreads several items	Frequently misinterprets questions

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ASSESSMENT TOOL 3: MCQ Test — Questions & Answers

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Questions

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- B. API access
- C. Platform console access
- D. Paper-based access

Answer Key

Q.No	Answer	Correct Option Text
1	B	Block storage
2	B	REST
3	B	Running code without managing servers directly
4	B	Reducing latency near data sources
5	B	Introduces an intermediate distributed layer between edge and cloud
6	A	File storage
7	B	Web APIs
8	B	Storage Area Network
9	B	Pay-per-use execution model
10	A	Shared access to services and documents
11	A	Managed application runtime
12	B	Large unstructured cloud data such as images and backups
13	B	Web API
14	A	Edge computing
15	D	Paper-based access

Explanations

Q	Why this is correct
1	Block storage presents raw, fixed-size blocks that a hypervisor formats into a virtual disk (VMDK/VHD) — the low-level, high-IOPS access VMs need. Object storage lacks a file system for boot volumes; file storage adds overhead unsuited to disk I/O.
2	REST (Representational State Transfer) uses simple HTTP verbs (GET/POST/PUT/DELETE) and JSON, making it lightweight and the de-facto standard for cloud web APIs, vs. heavier SOAP/XML.
3	Serverless (FaaS) still runs on servers, but the provider handles provisioning, scaling, and patching — the developer only supplies code that runs in response to events.
4	Edge computing pushes processing to devices/gateways physically close to where data is generated, cutting round-trip time to a distant cloud data center.
5	Fog computing sits between edge devices and the centralized cloud, aggregating and pre-processing data from many edge nodes before sending summaries onward.
6	File storage organizes data in a familiar folder/file hierarchy accessed via protocols like SMB/NFS — ideal for shared drives used by departments.
7	Browsers and clients interact with cloud backends through Web APIs (typically REST over HTTPS), not low-level system calls or manual tooling.
8	SAN = Storage Area Network, a dedicated high-speed network giving block-level access to consolidated storage devices.

Q	Why this is correct
9	Serverless bills per invocation/execution time (pay-per-use) rather than for pre-provisioned, always-on servers — a major cost benefit.
10	Cloud platforms centralize documents and services so distributed teams can access and edit the same resources concurrently.
11	A Platform-as-a-Service (PaaS) offering — e.g., a managed application runtime/database — lets developers deploy code without managing the underlying OS or hardware.
12	Object storage (e.g., S3-style) stores data as objects with metadata, scaling cheaply for unstructured content like images, video, and backups.
13	Web APIs expose defined endpoints/contracts that let applications call cloud services programmatically.
14	Smart factories need millisecond decisions, so processing at the edge (on-site gateways/controllers) avoids the latency of a round trip to the cloud.
15	Paper-based access is manual and outside digital cloud access patterns; browser, API, and console access are all standard cloud access methods.